

A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-23280410})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K , the normalized generator σ_x , the pair (a', J) representing the torsion class e_j , the auxiliary divisor I' , the compactly represented element t , a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \text{div}(t) + i_x(J) = 0, \quad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$ -basis of the ideal. For example, the ideal $J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}3306 + \mathbf{Z}(922 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\text{Cl}(K)$ (as HNF ideals). A character x on $\text{Cl}(K)/5$ is recorded by its value vector $(x(\bar{\kappa}_1), x(\bar{\kappa}_2), x(\bar{\kappa}_3))$; thus $(1, 0, 0)$ is the character dual to $\bar{\kappa}_1$. The column number j records which basis element e_j of $\text{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x . Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x .

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 + 23280410 = 0$, of discriminant -93121640 . Class group invariants $(90 \ 10 \ 10)$, class number 9000; the three stored generator ideals (HNF) are $\begin{pmatrix} 110 & 10 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 2783 & 2243 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5313 & 3002 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a , column e_1

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1223)y^3 + (31s - 54702)y^2 + (-262s - 77374)y + (5362s + 13916100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2442y^8 - 104512y^7 + 24840199y^6 - 1281442632y^5 \\ & + 37697310058y^4 - 653399852384y^3 + 7821002817556y^2 - 67564177 \\ & 254880y + 862993931458040 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944} \right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $92583348642112 = 2^6 \cdot 373 \cdot 3878323921$:

$$\begin{pmatrix} 2893229645066 & 676961525184 & 589502920126 & 52456742812 & 1664024087210 & 1604042756432 & 2768392970129 & 1370328784183 & 286764359040 & 318371394619 \\ 0 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 331 factors with signed exponents of absolute value up to 6678860368142069536232957063; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 1 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 2: character a , column e_2

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1223)y^3 + (31s - 54702)y^2 + (-262s - 77374)y + (5362s + 13916100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2442y^8 - 104512y^7 + 24840199y^6 - 1281442632y^5 \\ & + 37697310058y^4 - 653399852384y^3 + 7821002817556y^2 - 67564177 \\ & 254880y + 862993931458040 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2 = 2$:

$$\begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 335 factors with signed exponents of absolute value up to 8630016995894005890277435997; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 3: character a , column e_3

Prescribed character vector $(1 \ 0 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 + (-s - 1223)y^3 + (31s - 54702)y^2 + (-262s - 77374)y + (5362s + 13916100)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2442y^8 - 104512y^7 + 24840199y^6 - 1281442632y^5 \\ & + 37697310058y^4 - 653399852384y^3 + 7821002817556y^2 - 67564177 \\ & 254880y + 862993931458040 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ -1 \ 0 \ 1 \ 0 \ -1 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $70723915378000 = 2^4 \cdot 5^3 \cdot 83 \cdot 317 \cdot 349 \cdot 3851$:

$$\begin{pmatrix} 353619576890 & 226223787320 & 304845758910 & 184862879596 & 184632968166 & 181072105461 & 294212362154 & 138662390884 & 2477451635 & 239748746535 \\ 0 & 10 & 0 & 2 & 9 & 5 & 2 & 1 & 8 & 3 \\ 0 & 0 & 10 & 4 & 1 & 5 & 4 & 5 & 1 & 8 \\ 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 335 factors with signed exponents of absolute value up to 11360477318744101310774891592; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_3) = (0 \ 3 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 4: character b , column e_1

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 1386y^3 + (8s + 3762)y^2 + (596s - 3739335)y + (-540s + 9732000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2768y^8 + 13068y^7 - 5572722y^6 + 23993076y^5 + 1 \\ & 1830607504y^4 + 166890129220y^3 + 22124281186385y^2 - 87767550748 \\ & 800y + 101500391556000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944} \right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $8225582389200 = 2^4 \cdot 3^3 \cdot 5^2 \cdot 11 \cdot 17^4 \cdot 829$:

$$\begin{pmatrix} 79061730 & 46549740 & 35701836 & 56876842 & 57146203 & 68486330 & 39987348 & 59418310 & 11589448 & 11429765 \\ 0 & 510 & 306 & 496 & 93 & 36 & 410 & 103 & 230 & 31 \\ 0 & 0 & 102 & 0 & 66 & 44 & 0 & 38 & 14 & 86 \\ 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 144 factors with signed exponents of absolute value up to 1194421; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \ 0 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 5: character b , column e_2

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 1386y^3 + (8s + 3762)y^2 + (596s - 3739335)y + (-540s + 9732000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 2768y^8 + 13068y^7 - 5572722y^6 + 23993076y^5 + 1 \\ & 1830607504y^4 + 166890129220y^3 + 22124281186385y^2 - 87767550748 \\ & 800y + 101500391556000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25897180844700 = 2^2 \cdot 3^3 \cdot 5^2 \cdot 11 \cdot 17^2 \cdot 3017159$:

$$\begin{pmatrix} 50778785970 & 9994992240 & 46721424228 & 50392711342 & 21352507294 & 8267213598 & 901181427 & 47146733096 & 28986564526 & 2818175337 \\ 0 & 255 & 44 & 145 & 235 & 185 & 10 & 30 & 254 & 33 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 144 factors with signed exponents of absolute value up to 150820; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (2 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 6: character b , column e_3

Prescribed character vector $(0 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 1386y^3 + (8s + 3762)y^2 + (596s - 3739335)y + (-540s + 9732000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 2768y^8 + 13068y^7 - 5572722y^6 + 23993076y^5 + 1830607504y^4 + 166890129220y^3 + 22124281186385y^2 - 87767550748800y + 101500391556000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $437679842688000 = 2^{10} \cdot 3^6 \cdot 5^3 \cdot 11 \cdot 653^2$:

$$\begin{pmatrix} 7757640 & 1175400 & 4513800 & 3608708 & 7472272 & 2541578 & 4206062 & 4256803 & 309334 & 5024847 \\ 0 & 117540 & 50040 & 36236 & 105080 & 63184 & 54798 & 53598 & 53436 & 57559 \\ 0 & 0 & 30 & 18 & 28 & 18 & 24 & 20 & 26 & 28 \\ 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 154 factors with signed exponents of absolute value up to 210093297; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 0 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 7: character c , column e_1

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1497y^3 - 6sy^2 + 2485196y + 5144s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2994y^8 + 7211401y^6 + 8278771584y^4 + 4739146009936y^2 + 616016782981760$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944}\right)s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $6315714281800 = 2^3 \cdot 5^2 \cdot 7^2 \cdot 11^2 \cdot 331 \cdot 16091$:

$$\begin{pmatrix} 4101113170 & 3067607340 & 1559075136 & 594349927 & 1415021108 & 1424640901 & 1723410576 & 3394723676 & 2276746911 & 3299080474 \\ 0 & 70 & 35 & 34 & 57 & 0 & 23 & 57 & 23 & 53 \\ 0 & 0 & 11 & 6 & 9 & 5 & 10 & 5 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 191 factors with signed exponents of absolute value up to 136699532599658633497440783230; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_1) = (1 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 8: character c , column e_2

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1497y^3 - 6sy^2 + 2485196y + 5144s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2994y^8 + 7211401y^6 + 8278771584y^4 + 4739146009936y^2 + 616016782981760$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $205986867915136 = 2^7 \cdot 7 \cdot 11^3 \cdot 97 \cdot 1780663$:

$$\begin{pmatrix} 1170379931336 & 644981247460 & 433569542132 & 559686401357 & 397452442684 & 697936890398 & 236541726330 & 669833112681 & 1059099272518 & 355516631038 \\ 0 & 44 & 32 & 19 & 38 & 40 & 26 & 36 & 38 & 31 \\ 0 & 0 & 2 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 191 factors with signed exponents of absolute value up to 42440090988473300209087976590; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of +1. Certified value: $D_x(e_2) = (0 \ 3 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 9: character c , column e_3

Prescribed character vector $(0 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1497y^3 - 6sy^2 + 2485196y + 5144s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2994y^8 + 7211401y^6 + 8278771584y^4 + 4739146009936y^2 + 616016782981760$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $34606612328200 = 2^3 \cdot 5^2 \cdot 11 \cdot 47 \cdot 2543 \cdot 131611$:

$$\begin{pmatrix} 1730330616410 & 665384873580 & 896179888394 & 1137553599693 & 1270429925962 & 336531032077 & 1119072287583 & 1727037198790 & 1448226960918 & 1678143938401 \\ 0 & 10 & 0 & 4 & 6 & 5 & 8 & 0 & 3 & 7 \\ 0 & 0 & 2 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 196 factors with signed exponents of absolute value up to 224982216132742352795028223699; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \ 2 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s + 274)y^3 + (32s + 112610)y^2 + (490s + 1782800)y + (8000s + 22827000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 549y^8 + 224672y^7 + 26695866y^6 + 1593744920y^5 \\ & + 60266274140y^4 + 1516591629600y^3 + 25828669141000y^2 + 2639103 \\ & 65600000y + 2011018169000000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ 1 \ -1 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944} \right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $11028989958892 = 2^2 \cdot 2757247489723$:

$$\begin{pmatrix} 5514494979446 & 2618256060158 & 3319464559781 & 1351164409893 & 4824009945099 & 2835652319873 & 5424819910396 & 5248600430587 & 3999676076568 & 815881425666 \\ 0 & 2 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 306 factors with signed exponents of absolute value up to 34053130; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \ 0 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s + 274)y^3 + (32s + 112610)y^2 + (490s + 1782800)y + (8000s + 22827000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 549y^8 + 224672y^7 + 26695866y^6 + 1593744920y^5 \\ & + 60266274140y^4 + 1516591629600y^3 + 25828669141000y^2 + 2639103 \\ & 65600000y + 2011018169000000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ 1 \ -1 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $200469146154680 = 2^3 \cdot 5 \cdot 13 \cdot 385517588759$:

$$\begin{pmatrix} 100234573077340 & 44871537816188 & 23327344337645 & 86790156234051 & 32157992542773 & 54258146731631 & 12392539743258 & 78105122436465 & 57741376764562 & 50378541817906 \\ 0 & 2 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 291 factors with signed exponents of absolute value up to 49101263; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \ 3 \ 4)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \ 1 \ 0)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - y^4 + (s + 274)y^3 + (32s + 112610)y^2 + (490s + 1782800)y + (8000s + 22827000)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 2y^9 + 549y^8 + 224672y^7 + 26695866y^6 + 1593744920y^5 \\ & + 60266274140y^4 + 1516591629600y^3 + 25828669141000y^2 + 2639103 \\ & 65600000y + 2011018169000000 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \ -1 \ 1 \ -1 \ 0 \ -1 \ -1 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $78673189270000 = 2^4 \cdot 5^4 \cdot 13^2 \cdot 46552183$:

$$\begin{pmatrix} 6051783790 & 4849650 & 3473044500 & 4549931290 & 981268594 & 6048767317 & 5118394763 & 269035266 & 5883573778 & 3151208575 \\ 0 & 130 & 70 & 10 & 115 & 85 & 125 & 80 & 99 & 54 \\ 0 & 0 & 10 & 0 & 1 & 1 & 4 & 5 & 0 & 8 \\ 0 & 0 & 0 & 10 & 4 & 5 & 0 & 5 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 292 factors with signed exponents of absolute value up to 80679732; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 0 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 2431y^3 + (-20s - 41720)y^2 + (560s - 586034)y + (37440s + 55718936)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 4858y^8 - 73716y^7 + 4904573y^6 + 316624648y^5 + \\ & 13679143964y^4 - 743487973872y^3 - 31869757610684y^2 + 9109063945 \\ & 68352y + 35737998355948096 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944} \right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $23328435046648 = 2^3 \cdot 29 \cdot 37 \cdot 2717664847$:

$$\begin{pmatrix} 5832108761662 & 5040872477075 & 4399817404695 & 3433385733532 & 2880739741715 & 3411967324992 & 4660159833363 & 2991839361562 & 258514776876 & 3990591178780 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 184 factors with signed exponents of absolute value up to 10690372358; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 2 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 2431y^3 + (-20s - 41720)y^2 + (560s - 586034)y + (37440s + 55718936)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$\begin{aligned} & y^{10} - 4y^9 - 4858y^8 - 73716y^7 + 4904573y^6 + 316624648y^5 + \\ & 13679143964y^4 - 743487973872y^3 - 31869757610684y^2 + 9109063945 \\ & 68352y + 35737998355948096 \end{aligned}$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right)s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a')^J = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $34221031829920 = 2^5 \cdot 5 \cdot 13 \cdot 37 \cdot 4447 \cdot 99991$:

$$\begin{pmatrix} 2138814489370 & 1788347725198 & 724733819226 & 739031695364 & 1586823100520 & 1559220199762 & 2024767834750 & 1257104707363 & 1835324174168 & 399714773945 \\ 0 & 2 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 179 factors with signed exponents of absolute value up to 5330784735; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \ 1 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \ 0 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 - 2y^4 - 2431y^3 + (-20s - 41720)y^2 + (560s - 586034)y + (37440s + 55718936)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} - 4y^9 - 4858y^8 - 73716y^7 + 4904573y^6 + 316624648y^5 + 13679143964y^4 - 743487973872y^3 - 31869757610684y^2 + 910906394568352y + 35737998355948096$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \ 0 \ 0 \ 0 \ 1 \ 1 \ 1 \ -1 \ -1 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $322813702177280 = 2^9 \cdot 5 \cdot 37 \cdot 389 \cdot 8761141$:

$$\begin{pmatrix} 5043964096520 & 517972930926 & 4515589265244 & 601299924262 & 558944926565 & 2101125304032 & 47259213896 & 576745777600 & 1063922116709 & 4463379852256 \\ 0 & 2 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 4 & 0 & 3 & 2 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 183 factors with signed exponents of absolute value up to 390369661; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \ 4 \ 0)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1130y^3 - 20sy^2 - 1316000y + 8000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2260y^8 - 1355100y^6 + 6338004000y^4 - 5717875200000y^2 + 1489946240000000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{18989878}{12341395433243493} + \left(\frac{8299}{98731163465947944}\right) s, \quad J = \begin{pmatrix} 3306 & 922 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3306$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $16908048949088 = 2^5 \cdot 11^2 \cdot 59 \cdot 211 \cdot 350771$:

$$\begin{pmatrix} 96068459938 & 52202978938 & 31476526946 & 69117829246 & 43426832426 & 90147757002 & 37189300412 & 50445758877 & 86747348008 & 26830996644 \\ 0 & 22 & 21 & 6 & 13 & 18 & 18 & 19 & 10 & 16 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 310 factors with signed exponents of absolute value up to 496579981; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \ 2 \ 3)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1130y^3 - 20sy^2 - 1316000y + 8000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2260y^8 - 1355100y^6 + 6338004000y^4 - 5717875200000y^2 + 1489946240000000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{25968}{2704390420025} + \left(\frac{11}{108175616801000}\right) s, \quad J = \begin{pmatrix} 1610 & 610 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1610$. **Checked:** $(a') J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $267585311999500 = 2^2 \cdot 5^3 \cdot 11^2 \cdot 181 \cdot 24435899$:

$$\begin{pmatrix} 2432593745450 & 1569739141190 & 752875540384 & 1016810925366 & 28359707852 & 1169787635720 & 2039111369938 & 1600869988218 & 1699613782007 & 2193210113066 \\ 0 & 55 & 19 & 40 & 0 & 54 & 12 & 15 & 12 & 21 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 348 factors with signed exponents of absolute value up to 17422080028; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \ 4 \ 1)$ in the coordinates of $\text{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \ 1 \ 1)$, i.e. the values of x on the classes of the three stored generators of $\text{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K :

$$y^5 + 1130y^3 - 20sy^2 - 1316000y + 8000s$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

$$y^{10} + 2260y^8 - 1355100y^6 + 6338004000y^4 - 5717875200000y^2 + 1489946240000000$$

The automorphism σ_x is determined by the image of the primitive element θ : the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \ 2 \ 0 \ 1 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{26935527}{41838468347985914} + \left(-\frac{36035}{167353873391943656}\right)s, \quad J = \begin{pmatrix} 3674 & 1154 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3674$. **Checked:** $(a')J^5 = \mathcal{O}_K$, equivalently $\text{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $738812636320 = 2^5 \cdot 5 \cdot 11 \cdot 419779907$:

$$\begin{pmatrix} 92351579540 & 36730778152 & 1661939816 & 89598651484 & 80836332736 & 40356147804 & 88043681630 & 15037405572 & 18843641337 & 6603887778 \\ 0 & 4 & 3 & 1 & 3 & 2 & 2 & 2 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 343 factors with signed exponents of absolute value up to 26190595427; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \ 3 \ 2)$ in the coordinates of $\text{Cl}(K)/5$.